# RIG-FLATNESS DESCENDS ALONG FLAT BASE CHANGE AND FAITHFUL FLAT COVER

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the proof preserves the hypotheses under localization, reduces rig-flatness to naive rig-flatness, and descends flatness of the local maps along the faithfully flat surjective cover

*Editorial note.* The label lemma-rig-flat-local-down names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- faithful flatness characterized by flatness plus surjectivity of the induced map on spectra over an ideal of definition
- localization of adic rings at f and the corresponding local diagrams
- naive rig-flatness as the local form of rig-flatness, and descent of flatness along faithfully flat ring maps

**Lemma 0.1.** *Consider a commutative diagram*

$$\begin{array}{ccc} B & \longrightarrow & B' \\ {\scriptstyle\varphi}\uparrow & & \uparrow{\scriptstyle\varphi'} \\ A & \longrightarrow & A' \end{array}$$

*in $WAdm^{Noeth}$ with all arrows adic and topologically of finite type. Assume $A \to A'$ flat and $B \to B'$ faithfully flat. If $\varphi'$ is rig-flat, then $\varphi$ is rig-flat.*

*Proof.* Given $f \in A$ the assumptions of the lemma remain true for the digram

$$\begin{array}{ccc} B_{\{f\}} & \longrightarrow & (B')_{\{f\}} \\ {\scriptstyle\varphi}\uparrow & & \uparrow \\ A_{\{f\}} & \longrightarrow & (A')_{\{f\}} \end{array}$$

(To check the condition on faithful flatness: faithful flatness of $B \to B'$ is equivalent to $B \to B'$ being flat and $\mathrm{Spec}(B'/IB') \to \mathrm{Spec}(B/IB)$ being surjective for some ideal of definition $I \subset A$.) Hence it suffices to prove that $\varphi$ is naively rig-flat. However, we know that $\varphi'$ is naively rig-flat and that $\mathrm{Spec}(B') \to \mathrm{Spec}(B)$ is surjective. From this the result follows immediately. □

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